Kết quả validation (48 video)

Model	Threshold	APCER %	BPCER %	ACER %	EER %	AUC	Epochs	Best global
E0	0.990088	0.0000	0.0000	0.0000	0.0000	1.000000	5	5
E1	0.191604	0.0000	0.0000	0.0000	0.0000	1.000000	30	25
E2	0.411779	0.0000	0.0000	0.0000	0.0000	1.000000	10	1
E3	0.849619	0.0000	0.0000	0.0000	0.0000	1.000000	16	10
E3b	0.751813	0.0000	0.0000	0.0000	0.0000	1.000000	16	10
E4	0.418625	0.0000	0.0000	0.0000	0.0000	1.000000	16	14

CASIA development protocol · seed 42 · video mean of 20 frames · 270 attack / 90 live test videos. Fixed validation thresholds. No multi-seed uncertainty; E3 was selected before test.